

Forces and Moments:

The aerodynamic forces and moments on the body are due to only two basic sources:

1. Pressure distribution over the body surface
2. Shear stress distribution over the body surface

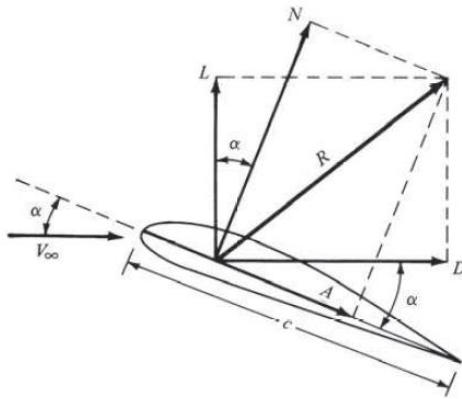


Fig.01 Resultant Aerodynamic Force

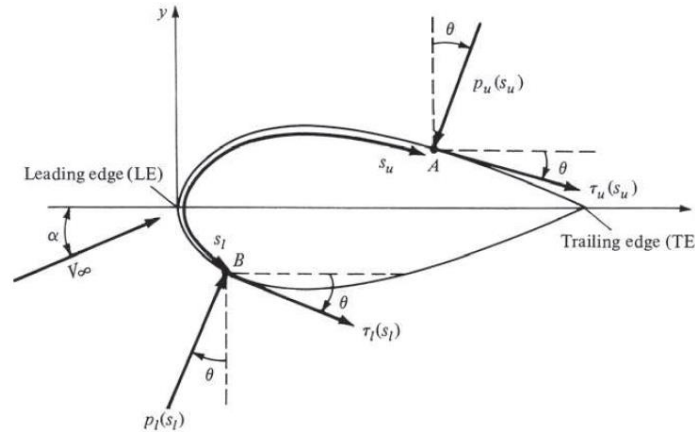


Fig.02 Pressure and shear stress over 2D body

N = Normal force (a component of **R** perpendicular to the chord)

A = Axial force (a component of **R** parallel to the chord)

L = Lift force (a component of **R** perpendicular to the freestream velocity)

D = Drag force (a component of **R** parallel to freestream velocity)

Resolve the pressure force and shear force along the x and y directions

Force due to pressure = $F_p = \text{Pressure} \times \text{Area}$

Force due to shear stress = $F_\tau = \text{Shear stress} \times \text{Area}$

Sign convention:

θ is positive when measured clockwise from the vertical line to the direction of **P** and τ .

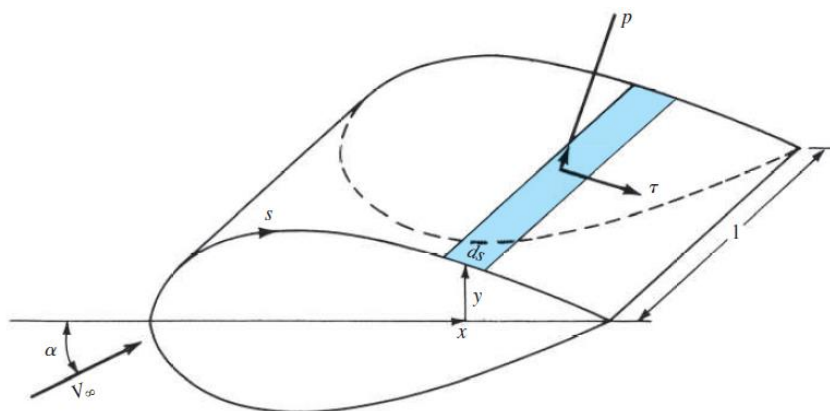


Fig.03 Aerodynamic force on an element of the body surface

Pressure and shear stress are the function of s (not a function of x, y). s coordinate follows the profile of the aerofoil and can represent a point on the profile.

x and y define the entire two-dimensional space and can define any point on that plane.

All the points on s can be represented by x and y coordinates.

$$s = f(x, y)$$

Elemental force on the upper body surface (per unit span):

$$dN_u = -P_u ds \cos \theta - \tau_u ds \sin \theta$$

$$dA_u = -P_u ds \sin \theta + \tau_u ds \cos \theta$$

Elemental force on the lower body surface (per unit span):

$$dN_l = P_l ds \cos \theta - \tau_l ds \sin \theta$$

$$dA_l = P_l ds \sin \theta + \tau_l ds \cos \theta$$

The total normal and axial forces *per unit span* are obtained by integrating from the leading edge (LE) to the trailing edge (TE):

$$N' = - \int_{LE}^{TE} (P_u \cos \theta + \tau_u \sin \theta) ds_u + \int_{LE}^{TE} (P_l \cos \theta - \tau_l \sin \theta) ds_l$$

$$A' = \int_{LE}^{TE} (-P_u \sin \theta + \tau_u \cos \theta) ds_u + \int_{LE}^{TE} (P_l \sin \theta + \tau_l \cos \theta) ds_l$$

$$M'_{LE} = \int_{LE}^{TE} [(P_u \cos \theta + \tau_u \sin \theta)x - (P_u \sin \theta - \tau_u \cos \theta)y] ds_u \\ + \int_{LE}^{TE} [(-P_l \cos \theta + \tau_l \sin \theta)x + (P_l \sin \theta + \tau_l \cos \theta)y] ds_l$$

Lift and Drag:

$$N' = - \int_{LE}^{TE} (P_u \cos \theta + \tau_u \sin \theta) ds_u + \int_{LE}^{TE} (P_l \cos \theta - \tau_l \sin \theta) ds_l$$

$$A' = \int_{LE}^{TE} (-P_u \sin \theta + \tau_u \cos \theta) ds_u + \int_{LE}^{TE} (P_l \sin \theta + \tau_l \cos \theta) ds_l$$

$$L = N \cos \alpha - A \sin \alpha$$

$$D = N \sin \alpha + A \cos \alpha$$

Reference:

Anderson, J. (2011): *Fundamentals of Aerodynamics (SI units)*. McGraw hill.